

RAVLIGHT.

Limitless Creations



VEYRON 1.0

PIXEL BAR

VEYRON 1.0

User and installation manual

Revision **B — 05 Oct 2026**

Firmware **2.24.0**

Contact **Ravision92 — Davide Ravenna · +39 346 018 2462**

Online **<https://ravlight.com/docs/fixtures/veyron.html>**

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This manual and the firmware

The tables in chapters 9, 10 and 11 are generated from firmware 2.24.0. A newer firmware can change personalities and macros: the current version, with animated macros, is always online at <https://ravlight.com/docs/fixtures/veyron.html>. The installed version is shown in the web interface under Settings.

1. Safety

Read before installing

Keep this manual and hand it to anyone who installs, operates or services the bar.

- Installation by qualified personnel only, following the electrical and safety rules of the venue.
- Disconnect power before connecting or disconnecting cables and before any maintenance.
- Power only from 100–240 V AC, 50/60 Hz, from an earthed line protected by a circuit breaker and a residual-current device.
- **Indoor use only (IP20).** Keep away from rain, moisture, condensation and dust.
- Suspended installations: use brackets rated for the weight (4.5 kg) and always add a secondary **safety cable**.
- The bar gets warm in use. Leave room for ventilation and do not cover it. Ambient temperature –10 °C to +45 °C.
- Do not stare into the sources at full power from close range.
- **Strobe:** strobe effects can trigger seizures in people with photosensitive epilepsy. Inform the audience when they are used.
- Do not open the bar. There are no user-serviceable parts inside.

Disposal

The product contains electronic components. At end of life take it to a WEEE collection point, not to household waste.

2. The bar

Veyron is a 1 m LED bar for console-controlled shows and installations. It has two independent sources:

- **RGB:** 40 pixels in two rows of 20, each one individually controllable.
- **White:** 3 white strips behind an optic of about 60°, divided into **6 segments**. Each segment has its own channel and lights the same section on all three strips.

It receives **Art-Net** and **sACN** over Ethernet or WiFi and **DMX512** over cable, with **RDM** on the DMX port. It is configured from a built-in web interface, with no software to install.

Front view and numbering

Seen from the front, **pixel 1 is top left** and the PowerCON input is on the right. The pixels form one serpentine chain: the top row runs 1 to 20 to the right, the bottom row comes back 21 to 40 to the left. Pixel 40 is therefore below pixel 1. The white segments W1–W6 are numbered from the left.

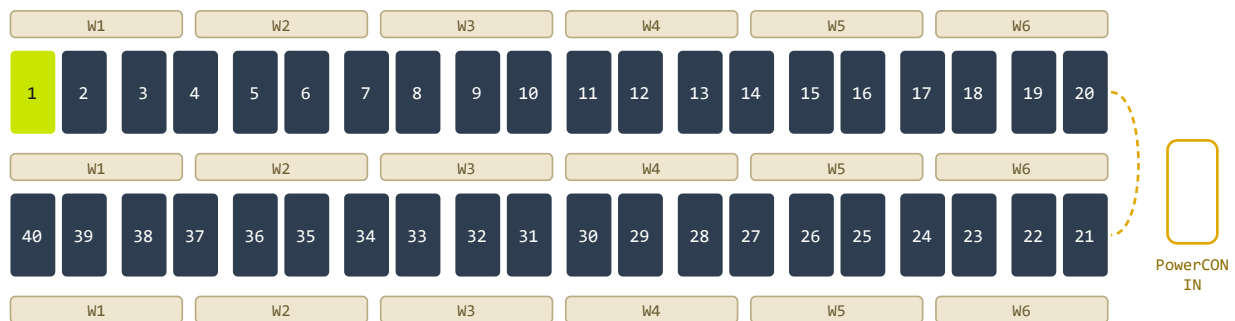


Figure 2.1 — Front view. Pixel 1 in green; the dashed line is where the chain passes from the top row to the bottom row.

Main features

- 9 DMX personalities, from 4 to 133 channels: single pixels, columns, groups of 2 pixels or the whole bar.
- Separate shutter and master dimmer for RGB and whites.
- 32 RGB macros that move over the console's colours, and 12 macros for the whites.
- Three independent DMX addresses (RGB, white and function blocks) and 4 dimming curves.
- Flip X / Flip Y orientation for bars hung upside down or mirrored.
- Runs without a console: recorded shows, built-in effects, signal-loss behaviour.
- Firmware updates over the network.

3. Specifications

Sources and optics

RGB	40 pixels in 2 rows × 20
White	3 white strips, 6 independent segments
White optic	about 60°
Theoretical output	Whites ≈ 12,000 lm · RGB ≈ 2,000–2,400 lm Nominal source values before the optic, not measured.

Electrical

Power supply	100–240 V AC, 50/60 Hz
Maximum power	200 W
Power linking	PowerCON IN / OUT, up to 10–12 bars per line (§5.1)

Control

Protocols	Art-Net, sACN (E1.31), DMX512, RDM
DMX channels	4 – 133 (9 personalities)
Universes	1 per bar
Network	Ethernet 10/100, WiFi 2.4 GHz
Configuration	Web interface, RDM
Firmware	2.24.0, updatable over the network

Mechanical and environmental

Dimensions	1015 × 83 × 90 mm Overall with connectors and brackets 1024 × 83 × 94.5 mm (§4)
Weight	4.5 kg
Protection	IP20, indoor only
Temperature	–10 °C ... +45 °C
Connectors	PowerCON IN/OUT, 3-pin XLR DMX IN/OUT, RJ45
In the box	PowerCON – Schuko cable, about 1.2 m

4. Mounting

The bar is mounted with 2.5 mm steel brackets. Always fix it to a structure that can carry its weight, and add a safety cable.

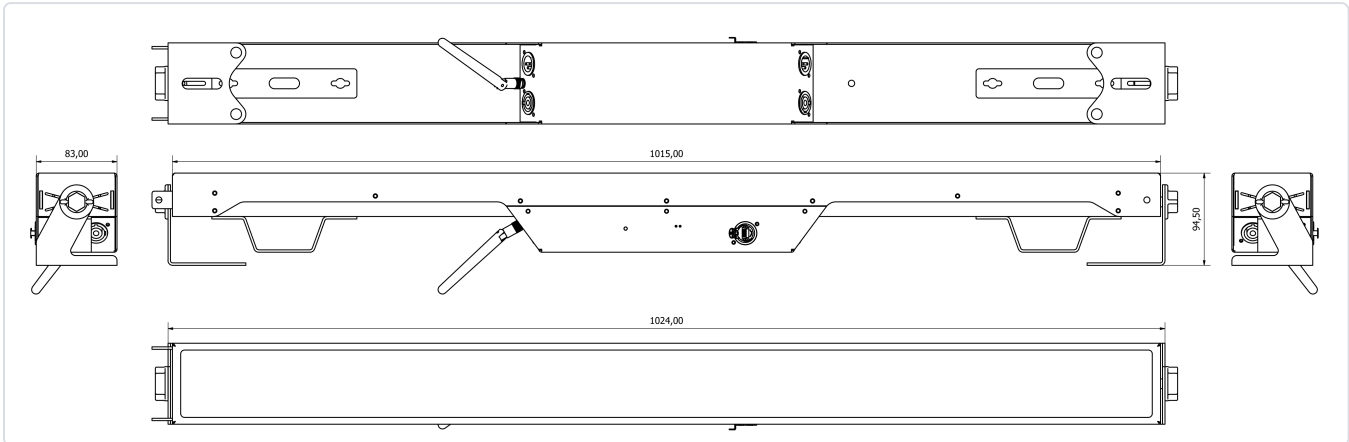


Figure 4.1 — Dimensioned views (mm): top with the bracket slots, rear, front, sides. Body 1015 mm, overall with connectors 1024 mm, height with brackets 94.5 mm.

- 1 Fix the bar with the brackets: to a wall or ceiling with screws through the slots, or to a truss with a clamp in the centre slot.
- 2 Tighten the screws and set the tilt with the knob.
- 3 Fit the safety cable between the bar and the structure (**attachment point**).
- 4 Point the WiFi antenna upwards if you use the wireless network.

5. Connections



Figure 5.1 — Rear of the bar: PowerCON and DMX connectors, Ethernet, WiFi antenna.

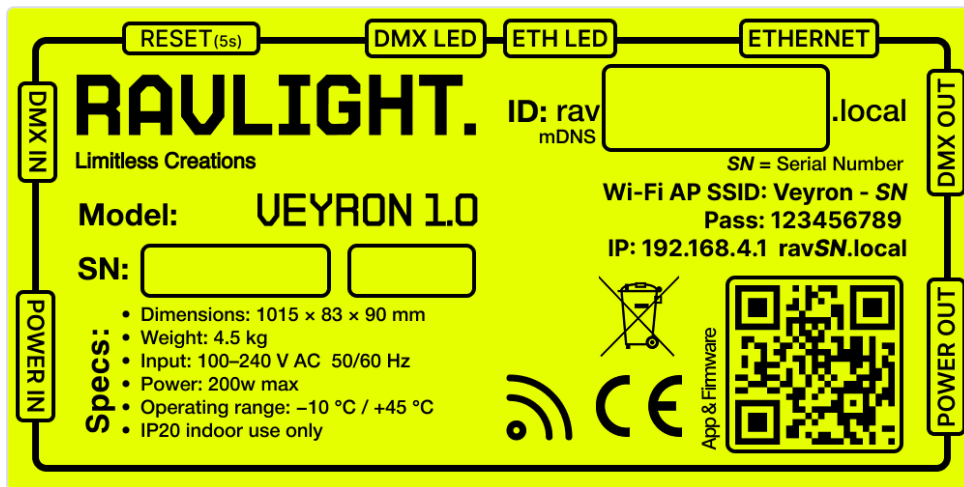


Figure 5.2 — Label: the edges mark POWER IN/OUT, DMX IN/OUT, Ethernet, RESET, DMX and ETH LEDs. In the middle: SN, access point credentials and a QR code for app and firmware.

5.1 Power

- Connect the first bar with the supplied PowerCON – Schuko cable. From **POWER OUT** you can feed the next bar with a PowerCON cable.
- One line comfortably carries **10–12 bars**: 12 bars at full load draw about 10–11 A at 230 V.
- Protect the line with a **16 A type C** circuit breaker and a 30 mA residual-current device. At switch-on the power supplies draw a current peak together, which can trip a type B breaker with too many bars on it.
- Do not plug or unplug PowerCON under load.

5.2 Data

Each bar receives **one DMX universe**. There are two ways to bring it:

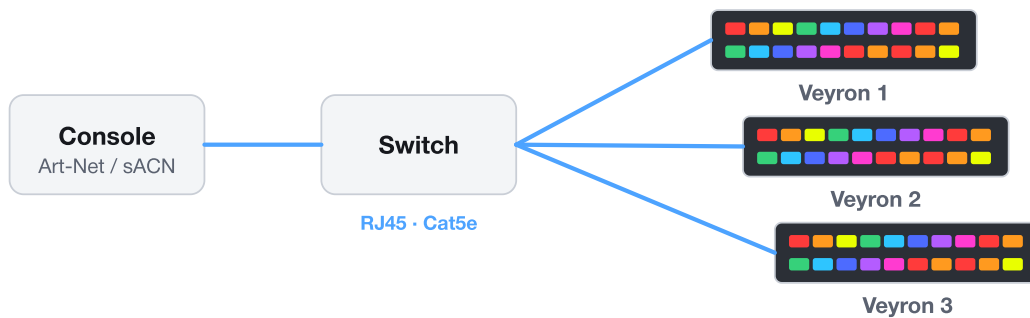
A — Network (recommended)

Each bar goes to a port on an Ethernet switch and the console sends Art-Net or sACN. Configuration and updates are done remotely on every bar.

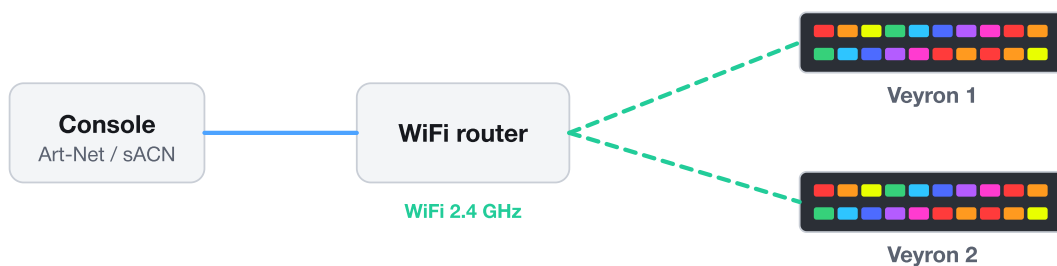
B — DMX cable

3-pin DMX cable from the console to **DMX IN**, then from **DMX OUT** to the next bar. Terminate the last bar with a 120 Ω terminator. Every bar in the chain reads the same universe.

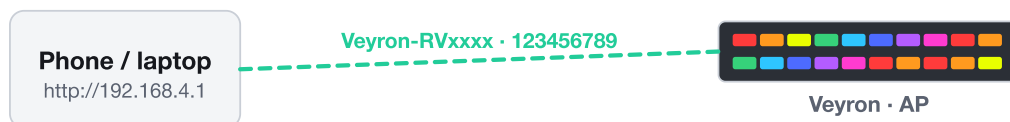
Wiring diagrams



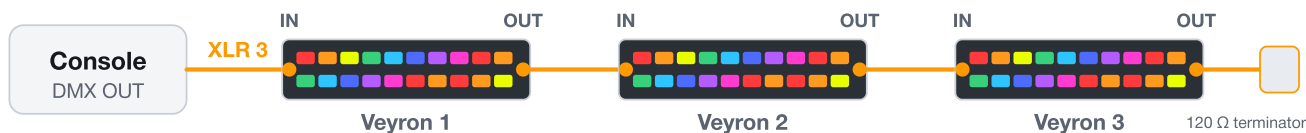
Ethernet network with a switch — Recommended. Each bar to a switch port; the console sends Art-Net or sACN. Configuration and updates remotely.



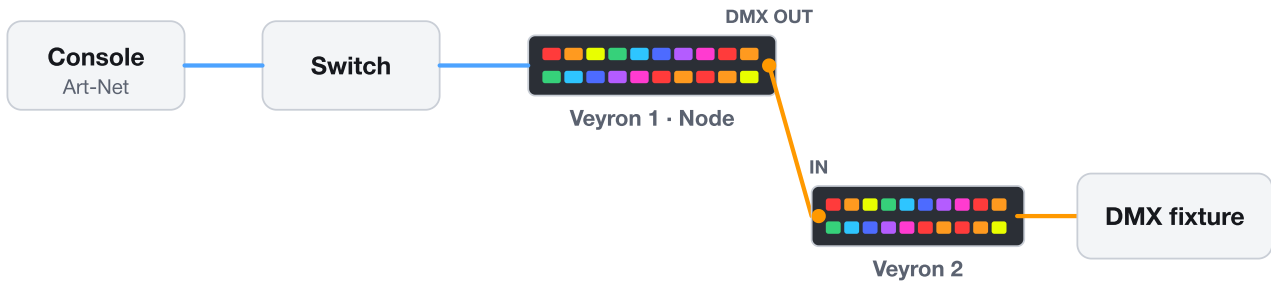
WiFi through a router — The bar joins the venue WiFi (Network section). Handy where cable cannot reach; for shows prefer Ethernet.



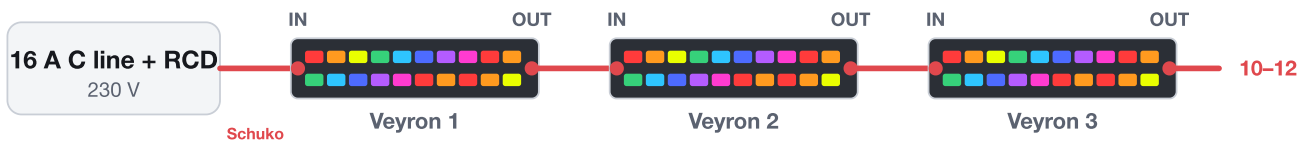
The bar's access point — With no network the bar opens its own WiFi. Use it for the first setup, from a phone or laptop.



DMX cable chain — 3-pin XLR from DMX IN to DMX OUT, 120 Ω terminator on the last bar. Every bar reads the same universe.



Bar as a DMX node — A networked bar repeats its universe on DMX OUT (DMX Node Output) and drives other bars or DMX fixtures over cable.



Power linking — PowerCON from IN to OUT, 10–12 bars per line. Line protected by a 16 A type C breaker and a 30 mA RCD.

DMX node

A bar that receives Art-Net or sACN can repeat its universe on **DMX OUT** (option *DMX Node Output*, §7). A networked bar can then drive other bars or traditional DMX fixtures over cable.

5.3 Addressing several bars

With the standard layout, a bar uses as many channels as its personality. The address of bar n on the same universe is:

$$\text{address}(n) = 1 + (n - 1) \times \text{personality channels}$$

Example with **P2 Full Pixel + Macro** (133 channels): bar 1 → 1, bar 2 → 134, bar 3 → 267. The fourth goes on another universe.

6. First start and network

Each bar has a unique serial number (**SN**) in the form `RVxxxx` (for example `RV1A2B`), printed on the label. It appears in the WiFi network name, the mDNS address and the web interface.

6.1 Connection sequence

- 1 Ethernet:** with the cable plugged in, the bar gets an address by DHCP (default) or uses the static IP you set.
- 2 WiFi:** without Ethernet, it tries the configured WiFi network.
- 3 Access point:** if neither works, it opens its own WiFi network.

While connecting, the bar shows two amber comets. When it gets an address it turns green for 1.5 s and then goes to DMX (§13).

6.2 Access point

Network (SSID)	Veyron-RVxxxx
Password	123456789
Address	<code>http://192.168.4.1</code>

Connect with a phone or laptop and open the address in a browser. The page usually opens by itself.

6.3 On a network

- By name: `http://ravRVxxxx.local` . Works only from the same network.
- By IP: the address given by DHCP, shown in the router or in the *Devices* panel of another bar.

6.4 If the network drops

- If the Ethernet cable comes out, the bar moves to the configured WiFi. After a few failed attempts it opens the access point and keeps retrying.
- With Ethernet connected, WiFi switches off after 10 s so it does not interfere.
- If a static IP is wrong, the bar goes back to DHCP by itself and restarts.

7. Web interface

The interface is split into sections. The *DMX active* indicator in the header lights while data arrives.

Network

Bar ID, Ethernet (DHCP or static IP), WiFi (network from the list, password, optional second network) and connection status. Network changes restart the bar: the page shows a countdown and reloads at the new address.

DMX

Source	Use
Art-Net (default)	Network. The bar appears by itself in compatible consoles and software.
sACN	Network, multicast per universe.
DMX Wired	Cable on DMX IN.
Auto Scene	Loops a recorded scene (§12).
Built-in Effects	Effects generated by the bar (§12).

Other fields: **Start Universe** (default 0), **DMX Node Output** (repeats the universe on DMX OUT, off by default) and signal-loss behaviour (§12).

Veyron

Personality, addresses, dimming curve, orientation, status LED and identify: see chapter 8.

Shows, Settings, Devices, DMX Monitor

- **Shows:** upload, play and record shows (§12).
- **Settings:** firmware version, update, restart, factory reset, configuration backup.
- **Devices:** the other RavLight units on the network.
- **DMX Monitor:** the 512 received channels live, with the bar's own channels highlighted. The quickest way to check a patch.

8. Fixture settings

Field	Default	Function
Highlight / Locate	—	Flashes the bar for 6 s so you can find it (§13).
RGB Pixel Start Address	1	First channel of the colour block.
White CW Pixel Start Address	121	First channel of the white block.
Function Start Address	127	First channel of shutter, master dimmer, macros and speed.
DMX Personality	P1	Channel layout (§9). The channel map of the chosen personality is shown below it.
Dimming Curve	Linear	Curve from DMX value to light, for RGB and whites.
Flip X / Flip Y	off	Orientation of the bar (§8.2).
Status LED	on	Shows network and update status on the bar (§13).

8.1 Three independent addresses

Each personality has three blocks, **RGB**, **White** and **Function**, each with its own address. When you change the RGB address or the personality, the interface suggests consecutive addresses for the other two, so the bar uses exactly the channels in the tables of chapter 10.

Console profiles

The Onyx and Avolites profiles assume the three blocks are consecutive. If you separate them, the console patch no longer matches the bar.

8.2 Orientation

Flip X	Flip Y	Result
—	—	Normal mounting: pixel 1 top left.
✓	—	Pixel 1 top right. Useful for symmetrical pairs of bars.
—	✓	Top and bottom rows swapped.
✓	✓	Bar hung upside down (PowerCON on the left).

The flip changes only the output to the LEDs. DMX channels, console profiles and pixel maps stay the same: the console always sees pixel 1 at the top left.

8.3 Dimming curves

Curve	Behaviour
Linear (default)	Output proportional to the DMX value.
Square	More detail at low values, smoother fades to black.
Inverse Square	A lot of light already at low values.
S Curve	Soft at both ends, fast through the middle.

9. DMX personalities

Each pixel layout comes as a **bare** version, with colour channels only, and a **+ Macro** version with shutter, master dimmer and macros. P1 is the original layout, kept unchanged so consoles already programmed against it keep working.

#	Personality	Channels	RGB strip	Whites	Function block	Bars per universe
1	Full Pixel (Legacy)	128	40 single pixels	6 segments	Shutter Strip, Shutter Accent	4
2	Full Pixel + Macro	133	40 single pixels	6 segments	Shutter Strip, Shutter Accent, Master Dimmer Strip, Master Dimmer Accent, Pixel Macro, White Macro, Macro Speed	3
3	Full Pixel	126	40 single pixels	6 segments	—	4
4	Mirror + Macro	73	20 columns (both rows)	6 segments	Shutter Strip, Shutter Accent, Master Dimmer Strip, Master Dimmer Accent, Zone Macro, White Macro, Macro Speed	7
5	Mirror	66	20 columns (both rows)	6 segments	—	7
6	Grouped 2px + Macro	73	20 zones of 2 pixels	6 segments	Shutter Strip, Shutter Accent, Master Dimmer Strip, Master Dimmer Accent, Zone Macro, White Macro, Macro Speed	7
7	Grouped 2px	66	20 zones of 2 pixels	6 segments	—	7
8	RGBW + Macro	10	whole bar, one colour	all together	Shutter Strip, Shutter Accent, Master Intensity, RGB Macro, White Macro, Macro Speed	51
9	RGBW	4	whole bar, one colour	all together	—	128

Which one to use

- **Full pixel mapping:** P2 Full Pixel + Macro, or P3 if you do not need shutter and macros.
- **Mirror** (P4, P5): 20 colour channels, one per column. Each column lights the top pixel and the one below it, so the bar behaves like a single row of 20.
- **Grouped 2px** (P6, P7): 20 zones of 2 adjacent pixels. Zones 1–10 on the top row from the left, zones 11–20 on the bottom row from the right.
- **The bar as one colour fixture:** P8 RGBW + Macro (10 channels) or P9 RGBW (4 channels).
- **Consoles already programmed with the original layout:** P1.

10. DMX tables

Channels with the three blocks consecutive from address 1: for another address add *address - 1*. The **Block** column gives the channel's position inside its block, useful when the blocks have separate addresses.

P1 — Full Pixel (Legacy) 128 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element	Channels	RGB +	Element
1 – 3	1	Pixel 1	61 – 63	61	Pixel 21
4 – 6	4	Pixel 2	64 – 66	64	Pixel 22
7 – 9	7	Pixel 3	67 – 69	67	Pixel 23
10 – 12	10	Pixel 4	70 – 72	70	Pixel 24
13 – 15	13	Pixel 5	73 – 75	73	Pixel 25
16 – 18	16	Pixel 6	76 – 78	76	Pixel 26
19 – 21	19	Pixel 7	79 – 81	79	Pixel 27
22 – 24	22	Pixel 8	82 – 84	82	Pixel 28
25 – 27	25	Pixel 9	85 – 87	85	Pixel 29
28 – 30	28	Pixel 10	88 – 90	88	Pixel 30
31 – 33	31	Pixel 11	91 – 93	91	Pixel 31
34 – 36	34	Pixel 12	94 – 96	94	Pixel 32
37 – 39	37	Pixel 13	97 – 99	97	Pixel 33
40 – 42	40	Pixel 14	100 – 102	100	Pixel 34
43 – 45	43	Pixel 15	103 – 105	103	Pixel 35
46 – 48	46	Pixel 16	106 – 108	106	Pixel 36
49 – 51	49	Pixel 17	109 – 111	109	Pixel 37
52 – 54	52	Pixel 18	112 – 114	112	Pixel 38
55 – 57	55	Pixel 19	115 – 117	115	Pixel 39
58 – 60	58	Pixel 20	118 – 120	118	Pixel 40

Channel	Block	Function	Default	Values
121	White +1	Accent White 1	0	0 – 255
122	White +2	Accent White 2	0	0 – 255
123	White +3	Accent White 3	0	0 – 255
124	White +4	Accent White 4	0	0 – 255
125	White +5	Accent White 5	0	0 – 255
126	White +6	Accent White 6	0	0 – 255
127	Function +1	Shutter Strip	0	RGB strip shutter · values
128	Function +2	Shutter Accent	0	White shutter · values

P2 — Full Pixel + Macro 133 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element
1 – 3	1	Pixel 1
4 – 6	4	Pixel 2
7 – 9	7	Pixel 3
10 – 12	10	Pixel 4
13 – 15	13	Pixel 5
16 – 18	16	Pixel 6
19 – 21	19	Pixel 7
22 – 24	22	Pixel 8
25 – 27	25	Pixel 9
28 – 30	28	Pixel 10
31 – 33	31	Pixel 11
34 – 36	34	Pixel 12
37 – 39	37	Pixel 13
40 – 42	40	Pixel 14
43 – 45	43	Pixel 15
46 – 48	46	Pixel 16
49 – 51	49	Pixel 17
52 – 54	52	Pixel 18
55 – 57	55	Pixel 19
58 – 60	58	Pixel 20

Channels	RGB +	Element
61 – 63	61	Pixel 21
64 – 66	64	Pixel 22
67 – 69	67	Pixel 23
70 – 72	70	Pixel 24
73 – 75	73	Pixel 25
76 – 78	76	Pixel 26
79 – 81	79	Pixel 27
82 – 84	82	Pixel 28
85 – 87	85	Pixel 29
88 – 90	88	Pixel 30
91 – 93	91	Pixel 31
94 – 96	94	Pixel 32
97 – 99	97	Pixel 33
100 – 102	100	Pixel 34
103 – 105	103	Pixel 35
106 – 108	106	Pixel 36
109 – 111	109	Pixel 37
112 – 114	112	Pixel 38
115 – 117	115	Pixel 39
118 – 120	118	Pixel 40

Channel	Block	Function	Default	Values
121	White +1	Accent White 1	0	0 – 255
122	White +2	Accent White 2	0	0 – 255
123	White +3	Accent White 3	0	0 – 255
124	White +4	Accent White 4	0	0 – 255
125	White +5	Accent White 5	0	0 – 255
126	White +6	Accent White 6	0	0 – 255
127	Function +1	Shutter Strip	0	RGB strip shutter · values
128	Function +2	Shutter Accent	0	White shutter · values
129	Function +3	Master Dimmer Strip	255	0 = off · 255 = full
130	Function +4	Master Dimmer Accent	255	0 = off · 255 = full
131	Function +5	Pixel Macro	0	RGB macro table
132	Function +6	White Macro	0	white macro table
133	Function +7	Macro Speed	0	0 = slow · 255 = fast

P3 — Full Pixel 126 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element
1 – 3	1	Pixel 1
4 – 6	4	Pixel 2
7 – 9	7	Pixel 3
10 – 12	10	Pixel 4
13 – 15	13	Pixel 5
16 – 18	16	Pixel 6
19 – 21	19	Pixel 7
22 – 24	22	Pixel 8
25 – 27	25	Pixel 9
28 – 30	28	Pixel 10
31 – 33	31	Pixel 11
34 – 36	34	Pixel 12
37 – 39	37	Pixel 13
40 – 42	40	Pixel 14
43 – 45	43	Pixel 15
46 – 48	46	Pixel 16
49 – 51	49	Pixel 17
52 – 54	52	Pixel 18
55 – 57	55	Pixel 19
58 – 60	58	Pixel 20

Channels	RGB +	Element
61 – 63	61	Pixel 21
64 – 66	64	Pixel 22
67 – 69	67	Pixel 23
70 – 72	70	Pixel 24
73 – 75	73	Pixel 25
76 – 78	76	Pixel 26
79 – 81	79	Pixel 27
82 – 84	82	Pixel 28
85 – 87	85	Pixel 29
88 – 90	88	Pixel 30
91 – 93	91	Pixel 31
94 – 96	94	Pixel 32
97 – 99	97	Pixel 33
100 – 102	100	Pixel 34
103 – 105	103	Pixel 35
106 – 108	106	Pixel 36
109 – 111	109	Pixel 37
112 – 114	112	Pixel 38
115 – 117	115	Pixel 39
118 – 120	118	Pixel 40

Channel	Block	Function	Default	Values
121	White +1	Accent White 1	0	0 – 255
122	White +2	Accent White 2	0	0 – 255
123	White +3	Accent White 3	0	0 – 255
124	White +4	Accent White 4	0	0 – 255
125	White +5	Accent White 5	0	0 – 255
126	White +6	Accent White 6	0	0 – 255

P4 — Mirror + Macro 73 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element
1 – 3	1	Column 1 pixel 1 + 40
4 – 6	4	Column 2 pixel 2 + 39
7 – 9	7	Column 3 pixel 3 + 38
10 – 12	10	Column 4 pixel 4 + 37
13 – 15	13	Column 5 pixel 5 + 36
16 – 18	16	Column 6 pixel 6 + 35
19 – 21	19	Column 7 pixel 7 + 34
22 – 24	22	Column 8 pixel 8 + 33

Channels	RGB +	Element
31 – 33	31	Column 11 pixel 11 + 30
34 – 36	34	Column 12 pixel 12 + 29
37 – 39	37	Column 13 pixel 13 + 28
40 – 42	40	Column 14 pixel 14 + 27
43 – 45	43	Column 15 pixel 15 + 26
46 – 48	46	Column 16 pixel 16 + 25
49 – 51	49	Column 17 pixel 17 + 24
52 – 54	52	Column 18 pixel 18 + 23

Channels	RGB +	Element
25 – 27	25	Column 9 pixel 9 + 32
28 – 30	28	Column 10 pixel 10 + 31

Channels	RGB +	Element
55 – 57	55	Column 19 pixel 19 + 22
58 – 60	58	Column 20 pixel 20 + 21

Channel	Block	Function	Default	Values
61	White +1	Accent White 1	0	0 – 255
62	White +2	Accent White 2	0	0 – 255
63	White +3	Accent White 3	0	0 – 255
64	White +4	Accent White 4	0	0 – 255
65	White +5	Accent White 5	0	0 – 255
66	White +6	Accent White 6	0	0 – 255
67	Function +1	Shutter Strip	0	RGB strip shutter · values
68	Function +2	Shutter Accent	0	White shutter · values
69	Function +3	Master Dimmer Strip	255	0 = off · 255 = full
70	Function +4	Master Dimmer Accent	255	0 = off · 255 = full
71	Function +5	Zone Macro	0	RGB macro table
72	Function +6	White Macro	0	white macro table
73	Function +7	Macro Speed	0	0 = slow · 255 = fast

P5 — Mirror 66 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element
1 – 3	1	Column 1 pixel 1 + 40
4 – 6	4	Column 2 pixel 2 + 39
7 – 9	7	Column 3 pixel 3 + 38
10 – 12	10	Column 4 pixel 4 + 37
13 – 15	13	Column 5 pixel 5 + 36
16 – 18	16	Column 6 pixel 6 + 35
19 – 21	19	Column 7 pixel 7 + 34
22 – 24	22	Column 8 pixel 8 + 33
25 – 27	25	Column 9 pixel 9 + 32
28 – 30	28	Column 10 pixel 10 + 31

Channels	RGB +	Element
31 – 33	31	Column 11 pixel 11 + 30
34 – 36	34	Column 12 pixel 12 + 29
37 – 39	37	Column 13 pixel 13 + 28
40 – 42	40	Column 14 pixel 14 + 27
43 – 45	43	Column 15 pixel 15 + 26
46 – 48	46	Column 16 pixel 16 + 25
49 – 51	49	Column 17 pixel 17 + 24
52 – 54	52	Column 18 pixel 18 + 23
55 – 57	55	Column 19 pixel 19 + 22
58 – 60	58	Column 20 pixel 20 + 21

Channel	Block	Function	Default	Values
61	White +1	Accent White 1	0	0 – 255
62	White +2	Accent White 2	0	0 – 255
63	White +3	Accent White 3	0	0 – 255
64	White +4	Accent White 4	0	0 – 255
65	White +5	Accent White 5	0	0 – 255

Channel	Block	Function	Default	Values
66	White +6	Accent White 6	0	0 – 255

P6 — Grouped 2px + Macro 73 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element	Channels	RGB +	Element
1 – 3	1	Zone 1 pixel 1 – 2	31 – 33	31	Zone 11 pixel 21 – 22
4 – 6	4	Zone 2 pixel 3 – 4	34 – 36	34	Zone 12 pixel 23 – 24
7 – 9	7	Zone 3 pixel 5 – 6	37 – 39	37	Zone 13 pixel 25 – 26
10 – 12	10	Zone 4 pixel 7 – 8	40 – 42	40	Zone 14 pixel 27 – 28
13 – 15	13	Zone 5 pixel 9 – 10	43 – 45	43	Zone 15 pixel 29 – 30
16 – 18	16	Zone 6 pixel 11 – 12	46 – 48	46	Zone 16 pixel 31 – 32
19 – 21	19	Zone 7 pixel 13 – 14	49 – 51	49	Zone 17 pixel 33 – 34
22 – 24	22	Zone 8 pixel 15 – 16	52 – 54	52	Zone 18 pixel 35 – 36
25 – 27	25	Zone 9 pixel 17 – 18	55 – 57	55	Zone 19 pixel 37 – 38
28 – 30	28	Zone 10 pixel 19 – 20	58 – 60	58	Zone 20 pixel 39 – 40

Channel	Block	Function	Default	Values
61	White +1	Accent White 1	0	0 – 255
62	White +2	Accent White 2	0	0 – 255
63	White +3	Accent White 3	0	0 – 255
64	White +4	Accent White 4	0	0 – 255
65	White +5	Accent White 5	0	0 – 255
66	White +6	Accent White 6	0	0 – 255
67	Function +1	Shutter Strip	0	RGB strip shutter · values
68	Function +2	Shutter Accent	0	White shutter · values
69	Function +3	Master Dimmer Strip	255	0 = off · 255 = full
70	Function +4	Master Dimmer Accent	255	0 = off · 255 = full
71	Function +5	Zone Macro	0	RGB macro table
72	Function +6	White Macro	0	white macro table
73	Function +7	Macro Speed	0	0 = slow · 255 = fast

P7 — Grouped 2px 66 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element	Channels	RGB +	Element
1 – 3	1	Zone 1 pixel 1 – 2	31 – 33	31	Zone 11 pixel 21 – 22
4 – 6	4	Zone 2 pixel 3 – 4	34 – 36	34	Zone 12 pixel 23 – 24
7 – 9	7	Zone 3 pixel 5 – 6	37 – 39	37	Zone 13 pixel 25 – 26
10 – 12	10	Zone 4 pixel 7 – 8	40 – 42	40	Zone 14 pixel 27 – 28

Channels	RGB +	Element	Channels	RGB +	Element
13 – 15	13	Zone 5 pixel 9 – 10	43 – 45	43	Zone 15 pixel 29 – 30
16 – 18	16	Zone 6 pixel 11 – 12	46 – 48	46	Zone 16 pixel 31 – 32
19 – 21	19	Zone 7 pixel 13 – 14	49 – 51	49	Zone 17 pixel 33 – 34
22 – 24	22	Zone 8 pixel 15 – 16	52 – 54	52	Zone 18 pixel 35 – 36
25 – 27	25	Zone 9 pixel 17 – 18	55 – 57	55	Zone 19 pixel 37 – 38
28 – 30	28	Zone 10 pixel 19 – 20	58 – 60	58	Zone 20 pixel 39 – 40

Channel	Block	Function	Default	Values
61	White +1	Accent White 1	0	0 – 255
62	White +2	Accent White 2	0	0 – 255
63	White +3	Accent White 3	0	0 – 255
64	White +4	Accent White 4	0	0 – 255
65	White +5	Accent White 5	0	0 – 255
66	White +6	Accent White 6	0	0 – 255

P8 — RGBW + Macro 10 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element
1 – 3	1	Strip Color all 40 pixels

Channel	Block	Function	Default	Values
4	White +1	Accent White	0	All 6 segments together · 0 – 255
5	Function +1	Shutter Strip	0	RGB strip shutter · values
6	Function +2	Shutter Accent	0	White shutter · values
7	Function +3	Master Intensity	255	Strip and whites together · 0 – 255
8	Function +4	RGB Macro	0	RGB macro table
9	Function +5	White Macro	0	white macro table
10	Function +6	Macro Speed	0	0 = slow · 255 = fast

P9 — RGBW 4 channels

RGB block — 3 channels per element, R · G · B order.

Channels	RGB +	Element
1 – 3	1	Strip Color all 40 pixels

Channel	Block	Function	Default	Values
4	White +1	Accent White	0	All 6 segments together · 0 – 255

11. Function channel values

The macros are shown animated on the wiki: <https://ravlight.com/docs/fixtures/veyron.html>.

11.1 Shutter Strip / Shutter Accent

One channel for RGB and one for the whites. The value picks the function and its speed together; speed rises with the value inside each band. Both ends are always Open, so a channel at 0 or 255 leaves the light on.

DMX value	Function	Effect
0 – 10	Open	Steady light.
11 – 70	Strobe	Short flash, slow to fast.
71 – 130	Random Strobe	Same flash, random interval on every cycle.
131 – 190	Pulse Open	Rises from dark to full and snaps closed. Slow to fast.
191 – 249	Pulse Close	Snaps on and fades to dark. Slow to fast.
250 – 255	Open	Steady light.

11.2 Master Dimmer / Master Intensity

Master Dimmer Strip and **Master Dimmer Accent** scale RGB and whites: 0 = off, 255 = full. The default is 255, so a console that does not use them still gets light. **Master Intensity** (P8 only) scales everything together.

11.3 Pixel Macro (P2) · Zone Macro (P4, P6) · RGB Macro (P8)

All three use the same set of 32 macros. Most **keep the console's colour**: they move a brightness mask over the colour of each pixel or zone. “Own colour” macros replace it, and “console + complement” macros blend between the console's colour and its complement. In P8 the console colour is Strip Color.

DMX value	Macro	Family	Colour	Description
0 – 9	No macro	—	console colour	The bar shows the colours sent by the console.
10 – 17	Chase Right	Sweeps	console colour	A bright head runs to the right with a 5-column tail. Both rows move together.
18 – 25	Chase Left	Sweeps	console colour	Like Chase Right, to the left.
26 – 33	Comet	Sweeps	console colour	Long tail with exponential decay: reads as one object crossing the bar.
34 – 40	Glitter Comet	Sweeps	console colour	Solid head, tail that sparkles pixel by pixel.
41 – 48	Scanner	Sweeps	console colour	A block bounces from one end to the other.
49 – 56	Wipe Right	Sweeps	console colour	Hard fill from the left, then the bar empties and starts again.
57 – 63	Wipe Left	Sweeps	console colour	Hard fill from the right.
64 – 71	Fade Right	Sweeps	console colour	Soft brightness wave travelling right. No hard edge: good for slow cues.
72 – 79	Fade Left	Sweeps	console colour	Soft wave travelling left.
80 – 86	Marquee	Sweeps	console colour	One column in three lit, stepping like theatre lights. Reads well from a distance.

DMX value	Macro	Family	Colour	Description
87 – 94	Dashes	Sweeps	console colour	Two-on, two-off blocks sliding along. Also works in Grouped 2px.
95 – 102	Iris In	Symmetry	console colour	Fill from both ends towards the centre.
103 – 109	Iris Out	Symmetry	console colour	Fill from the centre towards the ends.
110 – 117	Centre Pulse	Symmetry	console colour	A band at the centre that widens and narrows.
118 – 125	Split Chase	Symmetry	console colour	Two heads leave the centre together and run to the ends.
126 – 132	Serpentine Chase	Two rows	console colour	The head follows the wiring: right along the top row, left along the bottom row.
133 – 140	Row Alternate	Two rows	console colour	Top and bottom rows light in turn.
141 – 148	Counter Sweep	Two rows	console colour	Top row runs right, bottom row runs left: at speed it looks like a rotating drum.
149 – 156	Zipper	Two rows	console colour	Two heads enter from opposite ends on opposite rows and cross at the centre.
157 – 163	Rain Right	Two rows	console colour	Streaks running right along the bar, each with its own speed and random pause.
164 – 171	Rain Left	Two rows	console colour	Streaks running left.
172 – 179	Barber Pole	Two rows	console colour	A wave with the two rows offset: the bright band runs diagonally.
180 – 186	Twinkle	Organic	console colour	Random pixels flash to full on a dim base.
187 – 194	Sparkle Burst	Organic	console colour	Random sparks that fade out.
195 – 202	Noise Field	Organic	console colour	Slow, irregular breathing with no direction.
203 – 209	Fire	Organic	own colour	Fire with its own palette: the bottom row burns hotter.
210 – 217	Lightning	Organic	console colour	A long dark pause, then a double strike across the whole bar.
218 – 225	Breathe	Organic	console colour	The whole bar breathes together.
226 – 232	Rainbow Flow	Own colour	own colour	A full rainbow flowing along the bar.
233 – 240	Rainbow Split	Own colour	own colour	Rainbow with the rows half a cycle apart: top and bottom always show complementary colours.
241 – 248	Two-Tone Chase	Own colour	console + complement	Blocks alternating between the console colour and its complement, sliding.
249 – 255	Gradient Sweep	Own colour	console + complement	A moving blend between the console colour and its complement.

11.4 White Macro (P2, P4, P6, P8)

12 effects on the 6 white segments, independent of RGB.

DMX value	Macro	Description
0 – 9	No macro	The whites follow their own channels.
10 – 30	Breathe	All whites breathe together, down to full dark.
31 – 50	Chase Right	A segment runs right with a fading tail.
51 – 71	Chase Left	Like Chase Right, to the left.

DMX value	Macro	Description
72 – 91	Supercar	Back-and-forth scanner.
92 – 112	Twinkle	Random segments flash on.
113 – 132	Flicker	Irregular, flame-like flicker.
133 – 153	Fade Right	Soft wave travelling right.
154 – 173	Fade Left	Soft wave travelling left.
174 – 194	Slide Right	Fill from the left, then start again.
195 – 214	Slide Left	Fill from the right.
215 – 235	Mirror In	Fill from the ends towards the centre.
236 – 255	Mirror Out	Fill from the centre towards the ends.

11.5 Macro Speed

One channel sets the speed of every running macro: 0 = slowest, 255 = fastest.

12. Standalone operation

12.1 DMX signal loss

In the DMX section, under *When DMX stops arriving*, choose what the bar does when data stops:

Option	Behaviour
Hold (default)	Keeps the last received state.
Black	Turns the bar off.
Play a show	Starts a show stored on the bar. If the file is missing, it holds.
Run an effect	Starts a built-in effect.

- Delay: default 10 s, from 0 to 3600 s. 0 turns the feature off.
- It triggers only when every source (Art-Net, sACN, DMX cable) has stopped.
- With *give back when DMX returns* on (default), the console takes over again as soon as data comes back.
- The timer starts at power-up: a bar with no console waits the delay and then runs the action. Use this for installations that start by themselves.

12.2 Shows

- Uncompressed FSEQ v2, exported from xLights and similar software. Memory for about 2 minutes of show.
- Each show has play/stop and a *Loop until stopped* option.
- The bar can also **record** what it receives, up to 5 minutes. Record while preparing, not during the show: the bar can flicker while it writes.

12.3 Scene Recorder (Auto Scene)

4 scenes of 10 s recorded from the received universe. With the source set to **Auto Scene**, the chosen scene loops.

12.4 Built-in effects

With the source set to **Built-in Effects** the bar runs on its own: RGB and white macros, colour, speed, shutter and intensity are set from the interface.

13. Status and identify

13.1 Status on the bar

With *Status LED* on, the bar shows its status at about 10 % brightness and pauses DMX meanwhile.

State	What you see
Firmware update	 A blue bar filling with progress; the two centre white segments breathe.
Connecting	 Two amber comets, one per row, left to right.
Access point on	 Pixels 1 and 40 breathe magenta, with the four centre whites. Stops when a device connects.
Connected	 Green bar and outer whites for 1.5 s, then DMX.
Normal	No status shown: the bar follows DMX.

13.2 Rear LEDs

- **DMX LED:** breathes while DMX data arrives from any source. Goes off 1 s after the last data.
- **ETH LED:** on with the Ethernet cable connected.

13.3 Highlight / Locate

For 6 s two full-brightness white comets cross the bar and the centre whites breathe, then the bar returns to DMX. Start it from the *Highlight / Locate* button, the Devices panel or RDM Identify.

14. RDM

On the DMX port the bar answers RDM with a unique identifier. From the console you can:

- read and change the **personality**, all 9 with name and channel count;
- change the **address**: the three blocks move together, and the command is refused if a block would fall outside 1–512;
- read and write the **label**, which is the bar's ID;
- start **identify** (Highlight).

15. Updates, reset and maintenance

15.1 Firmware update

- 1 Download the Veyron firmware from <https://ravlight.com/docs/fixtures/veyron.html>.
- 2 Open *Settings* → *Update* in the web interface and upload the file.
- 3 The bar shows the progress in blue and restarts. Configuration and personality are kept.

Do not cut power during an update.

After an update, check on the wiki whether personalities or macros have changed.

15.2 Factory reset

- **RESET button:** hold it for **5 seconds**. The bar returns to factory settings and restarts. A short press does nothing.
- **Web interface:** *Settings* → *Reset*.

A reset erases network, addresses and personality: the bar goes back to DHCP, Art-Net universe 0, P1, address 1. Save a backup from Settings first.

15.3 Maintenance

- Clean the diffuser with a soft dry cloth, no solvents, with the bar off and cool.
- Check connectors and cables regularly.
- Keep the ventilation areas free of dust.

16. Troubleshooting

Problem	What to check
The bar does not respond to the console	In <i>DMX Monitor</i> check that data arrives on the right channels. Check source (Art-Net / sACN / Wired), universe and address. Some consoles number Art-Net universes from 1: the bar's universe 0 is their universe 1.
Colours or pixels shifted	Different personality or address between bar and console, or blocks that are not consecutive.
Pixel 1 is on the wrong side	The bar is hung upside down or turned: use Flip X / Flip Y (§8.2).
DMX LED breathes but the bar is dark	Master Dimmer at 0, shutter in a strobe or pulse band, or colour channels at zero.
Amber comets that never stop	The bar is looking for a network: check cable and switch. After a few attempts it opens the access point.
Pixels 1 and 40 magenta	Access point on: connect to <code>Veyron-RVxxxx</code> to set up the network.
Configuration lost	The RESET button was held for 5 s. Restore the backup from Settings.
Cannot find the IP	Try <code>ravRVxxxx.local</code> , the Devices panel of another bar or the router's DHCP list.
Breaker trips at switch-on	Too many bars on one line or a type B breaker: split the bars over more lines or use a 16 A type C (§5.1).

17. Product status and contact

Veyron is an experimental project. Conformity testing (low voltage, electromagnetic compatibility, radio equipment) has not been completed yet. Specifications may change without notice.

Contact	Ravision92 — Davide Ravenna
Phone	+39 346 018 2462
Documentation and firmware	https://ravlight.com/docs/fixtures/veyron.html

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